

A Resilient On-Chip STDP Learning Under Memory Voltage Scaling

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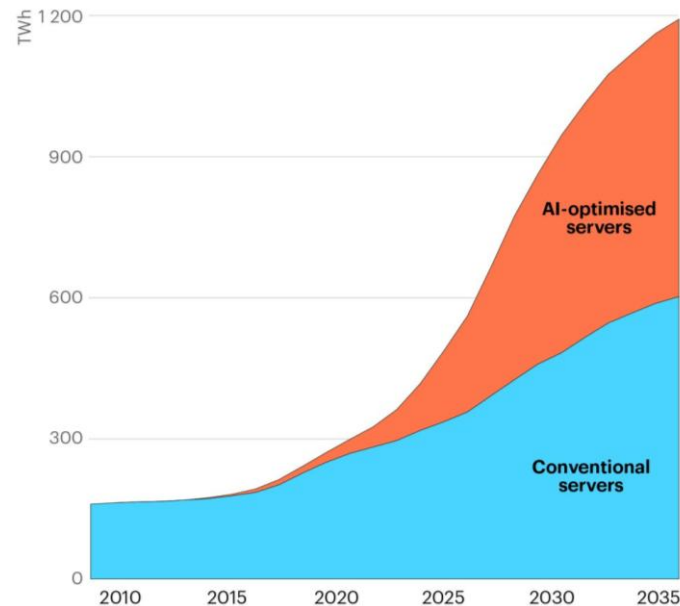
Content

- Introduction
- Proposed Method
- Results
- Conclusion

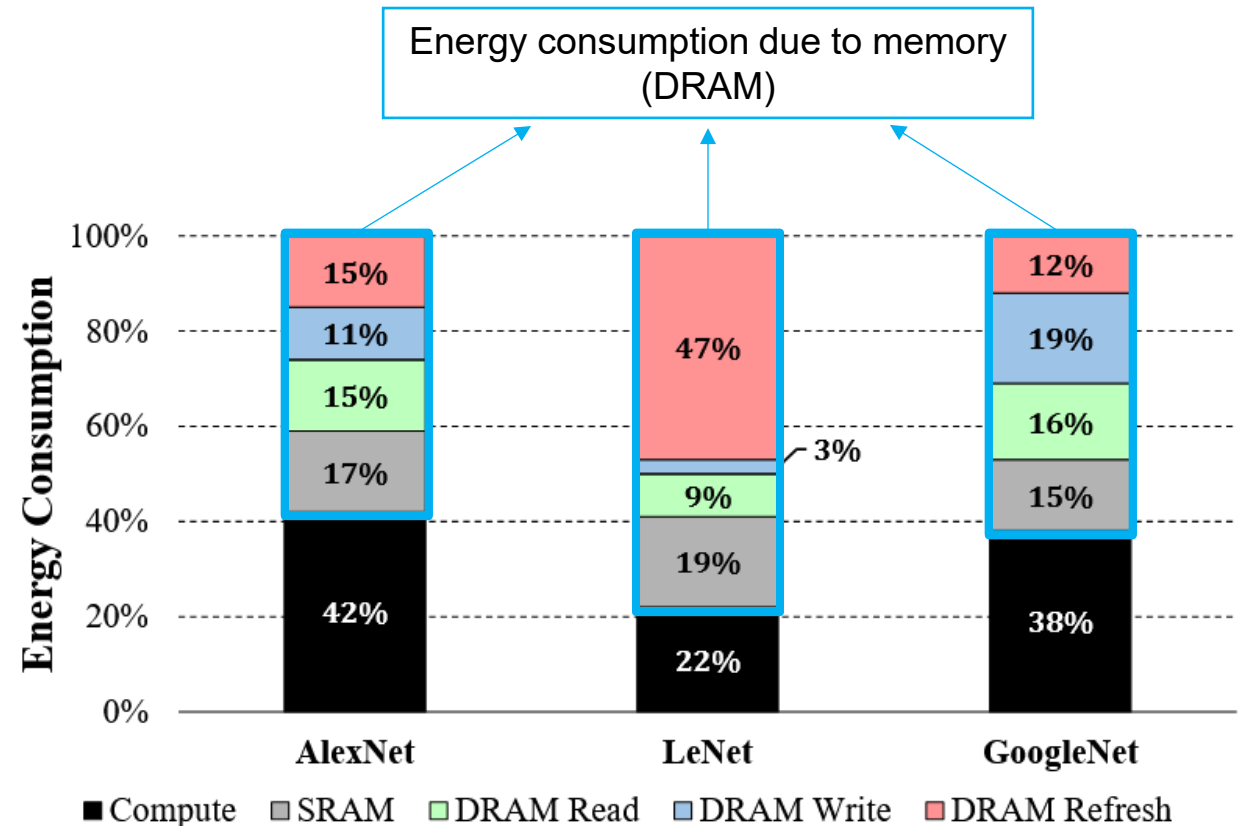
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 - Rate Coding
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 - STDP Learning
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Introduction

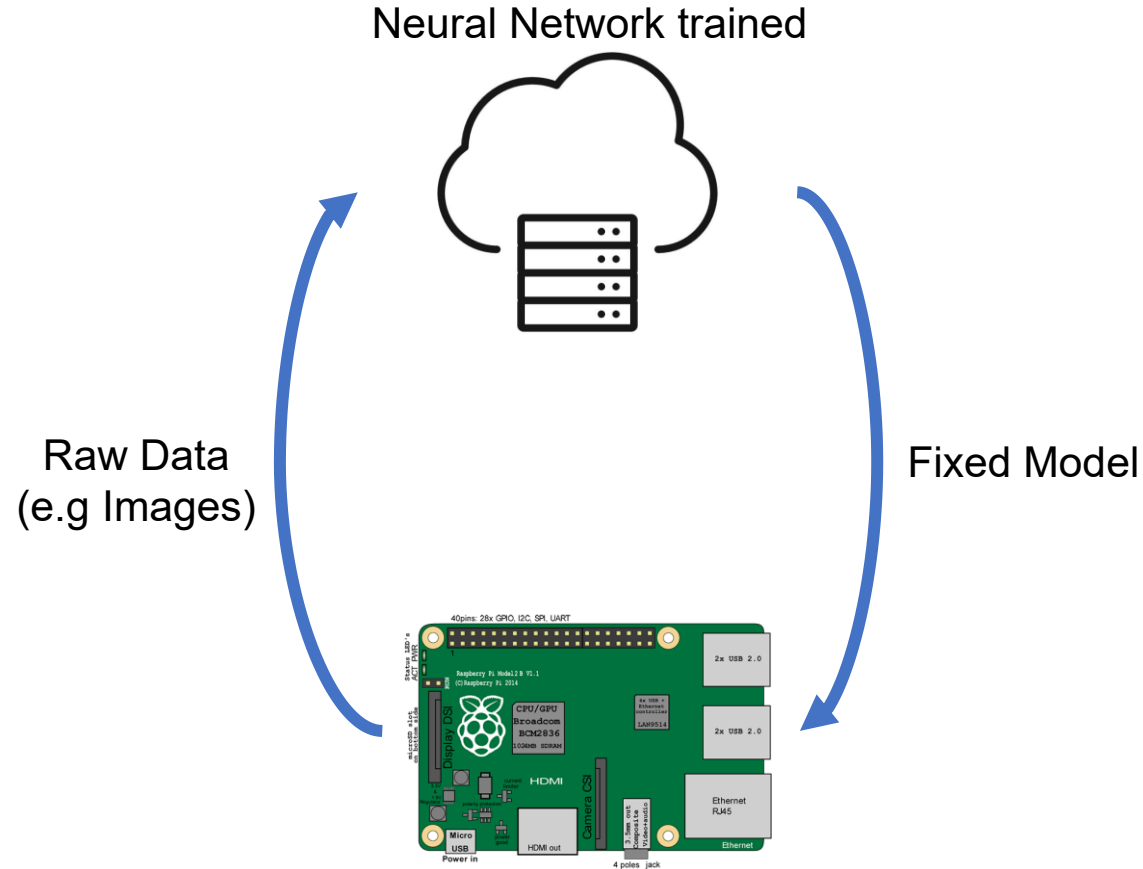


(a) Predicted power consumption due to AI usage ^[1]

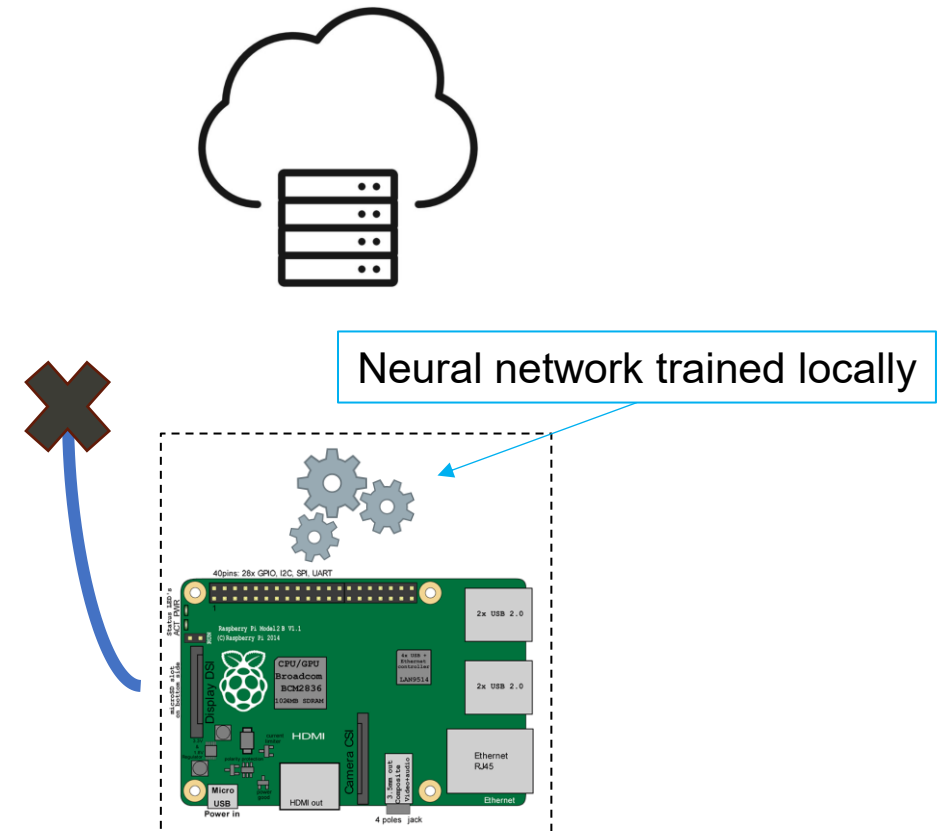


(b) Energy consumption breakdown of three CNNs on a modern CNN accelerator ^[2]

Off-Chip and On-Chip Learning

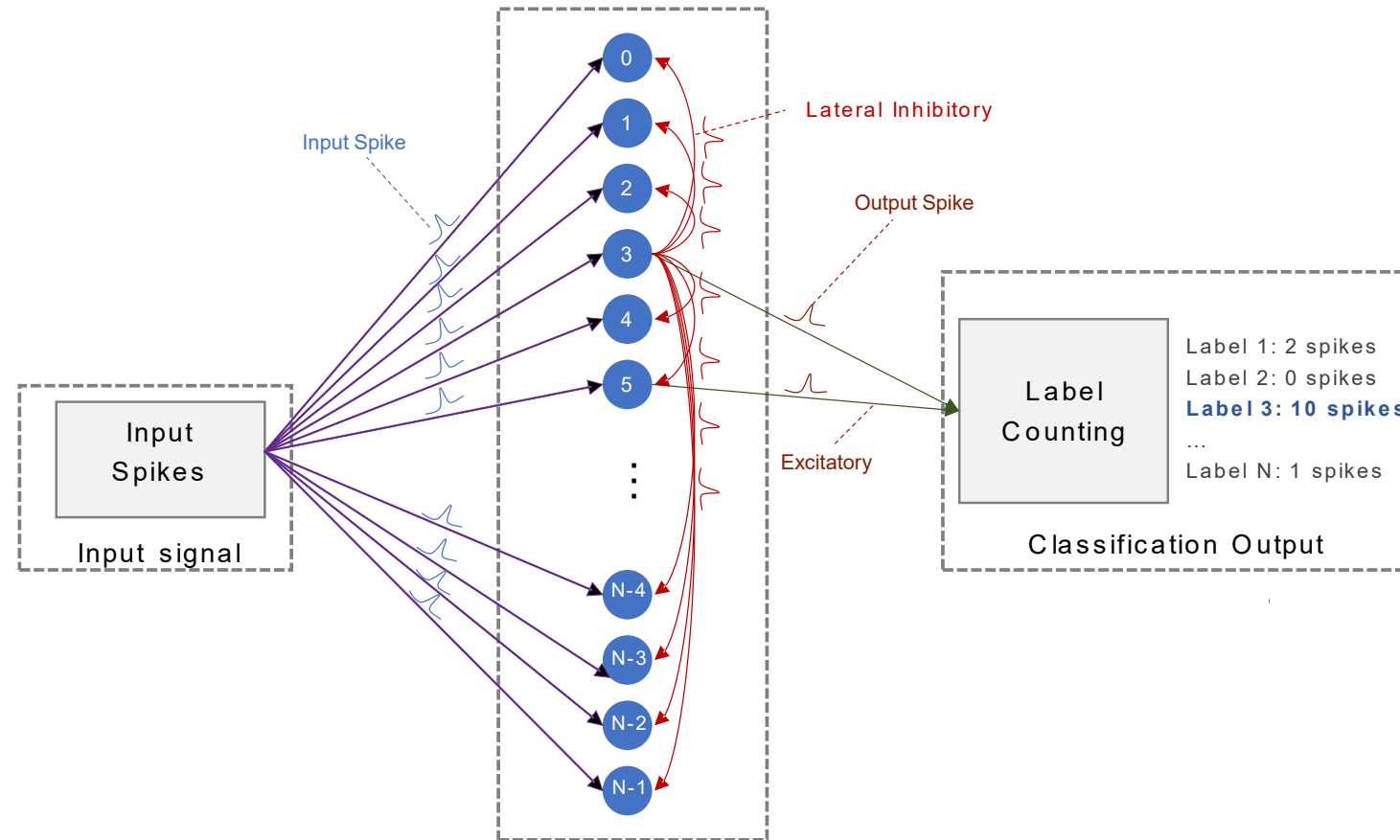


(a) Visual representation of Off-Chip learning



(b) Visual representation of On-Chip learning

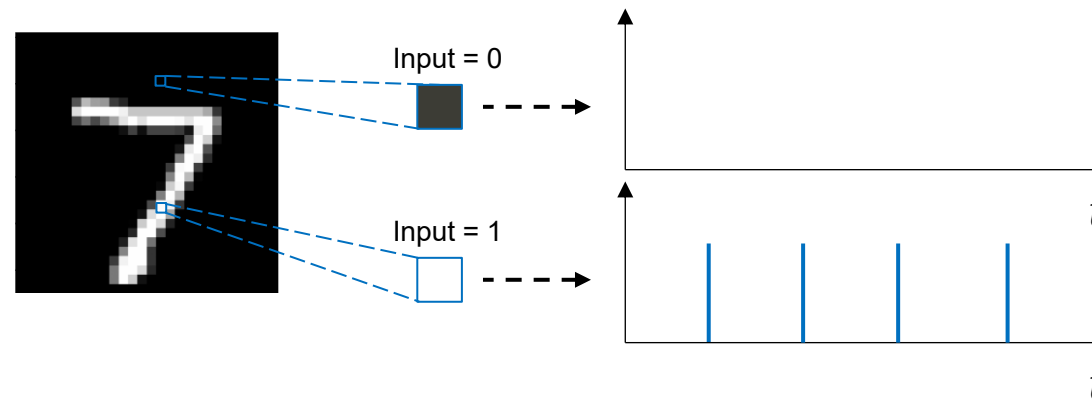
Spiking Neural Networks



A Spiking Neural Network (SNN) topology employing Spike-Timing Dependent Plasticity (STDP)

Rate Coding

- Converts raw sensor data into spike trains and processed by an SNN
- High pixel intensity = High spike frequency
- Transmitting 1-bit spikes is more efficient

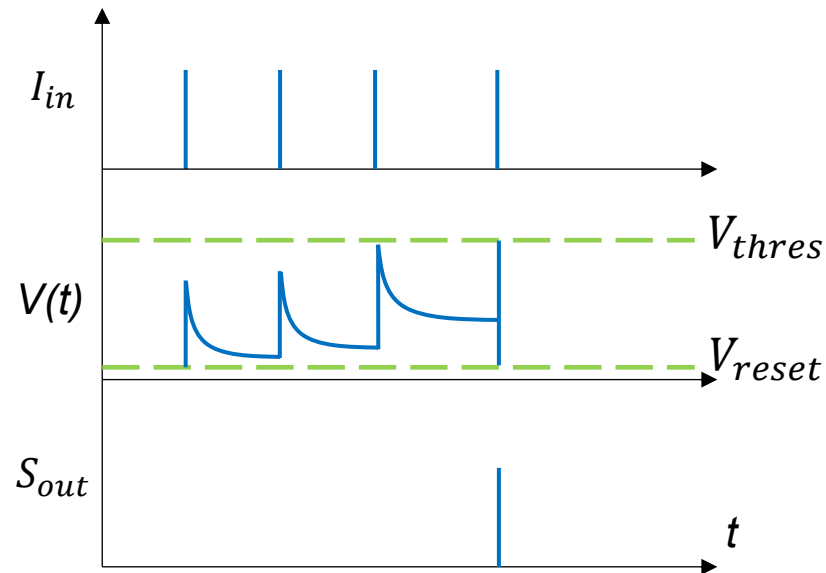


Spikes generated from MNIST image

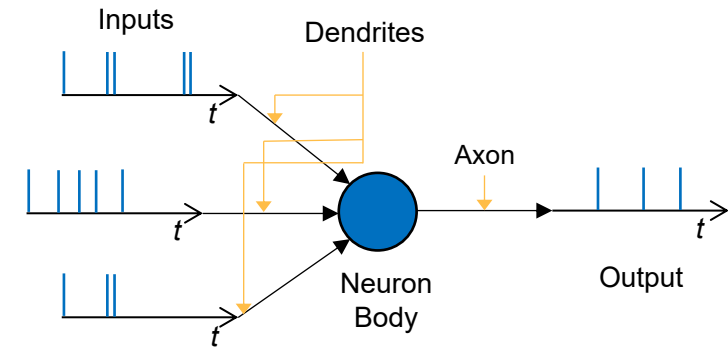
Leaky Integrate and Fire (LIF) Neuron

τ : Time constant
 $U(t)$: Membrane potential
 R : Membrane resistance
 I_{in} : Input current

S_{out} : Output spike
 V_{thres} : Threshold potential
 V_{reset} : Reset potential



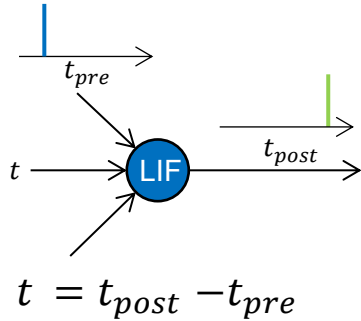
(a) LIF Potential dynamics



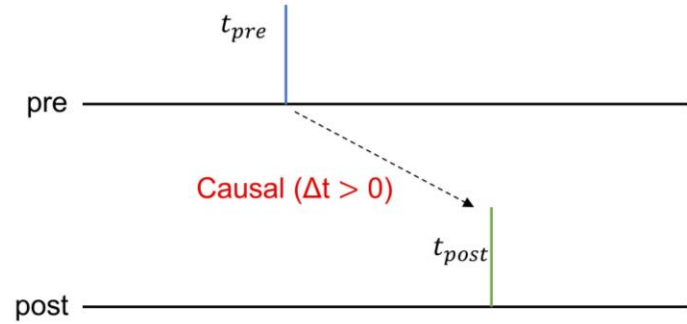
(b) Typical SNN Neuron

LIF equation:
$$\tau \frac{dU(t)}{dt} = -U(t) + RI_{in}(t)$$

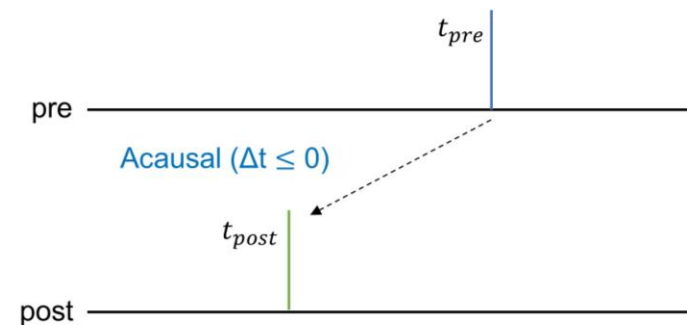
STDP Learning



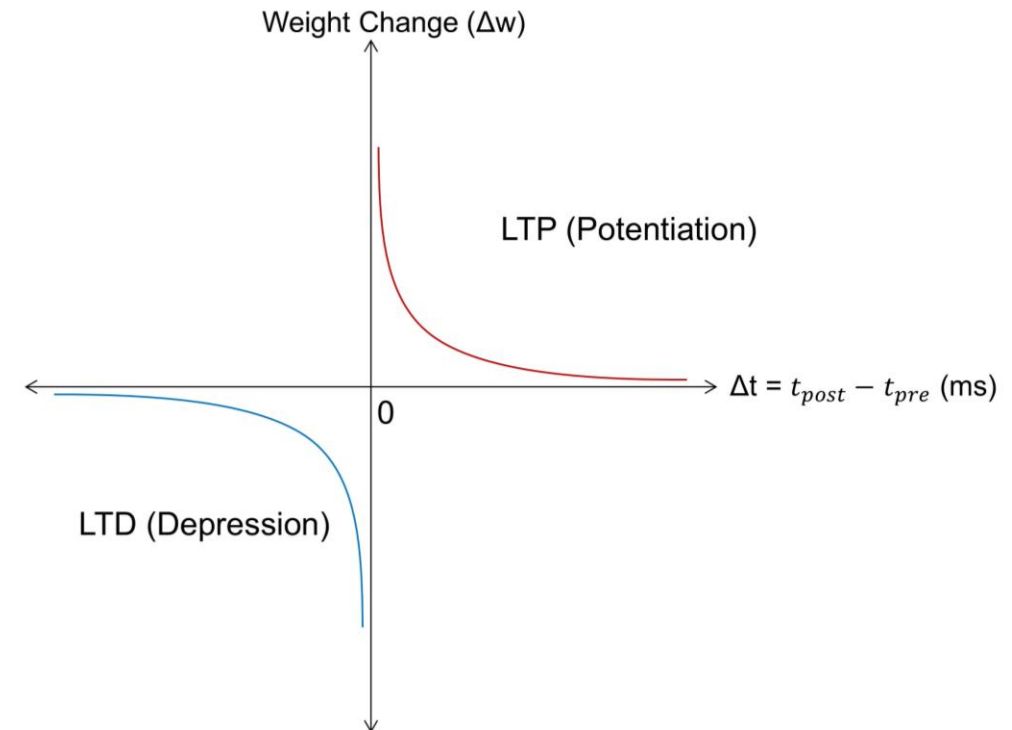
(a) LIF neuron and spike timings



(b) LTP Timing



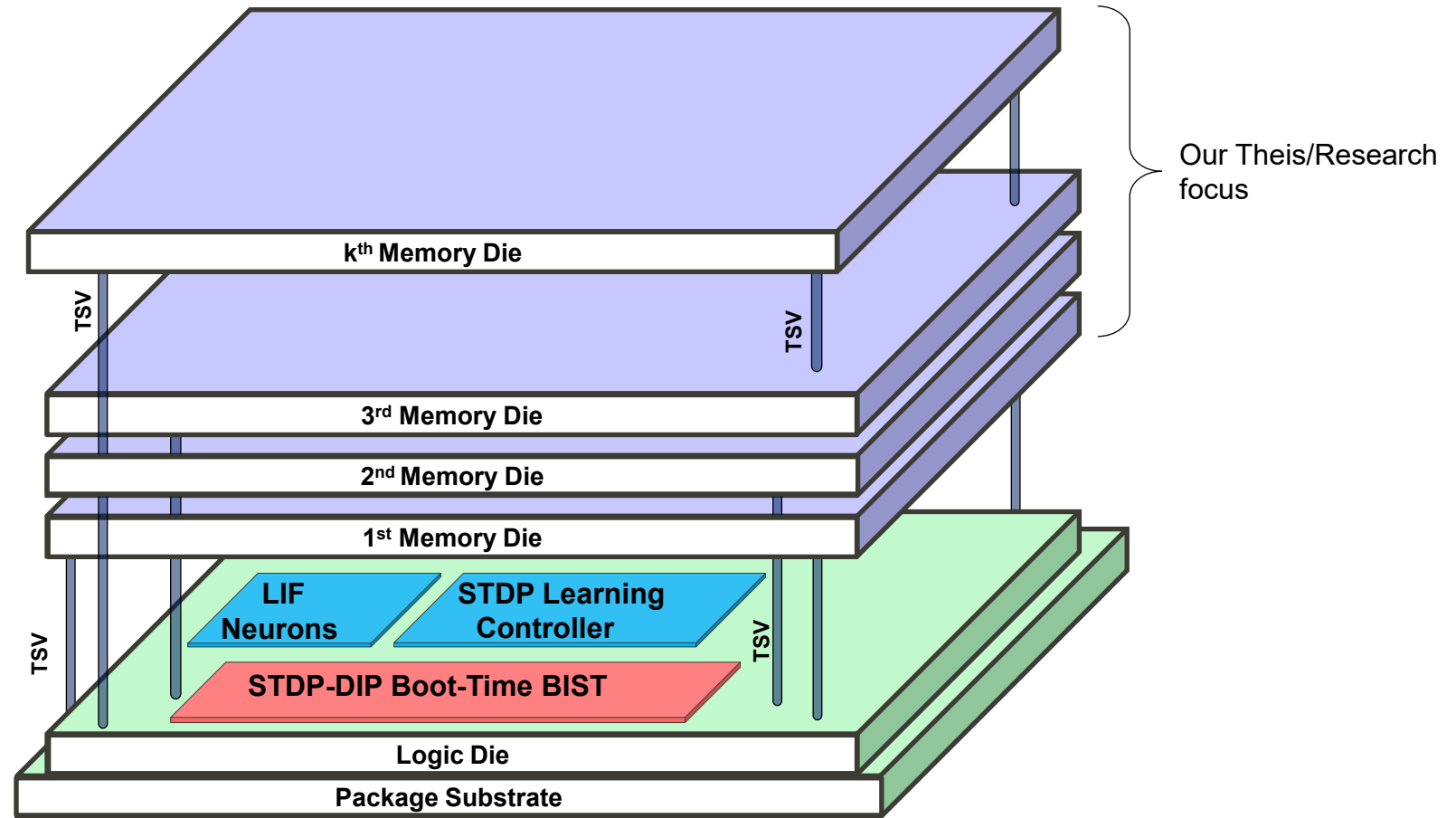
(c) LTD Timing



(d) STDP Learning Window

$$\text{Equation: } \Delta w = \eta (x_{pre} - x_{tar}) (w_{max} - w)^\mu$$

Overall Architecture



Architecture of proposed system

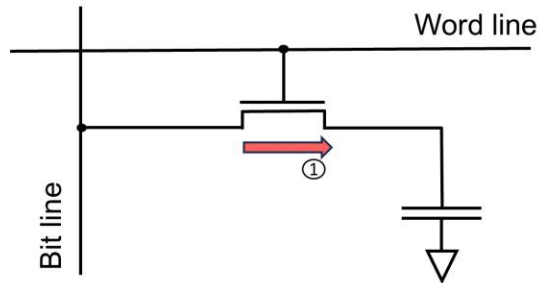
Part of "Y. Hanyu, R. Kobayashi, S. R. Hariprakash, S. Ganesh, N.-D. Nguyen, D.-A. Nguyen, M. Maatar, A. Ben Abdallah, Z. Wang, and K. N. Dang, 'E3STDP: Energy-Efficient Ensemble Spike-Timing-Dependent Plasticity for Carbon-Aware Neuromorphic Computation.'" under major revision

Wednesday, August 5, 2026

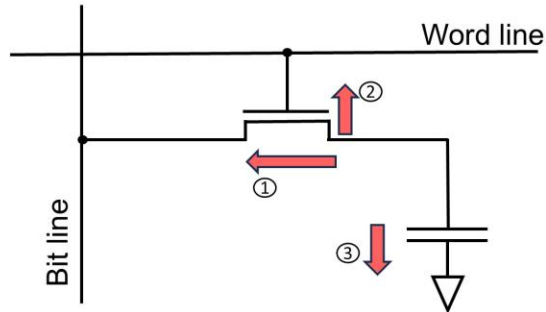
Master's Thesis | Subbaiah Ravi Hariprakash

Undervolting DRAM

① Subthreshold leakage ② Gate-induced drain leakage ③ Junction leakage

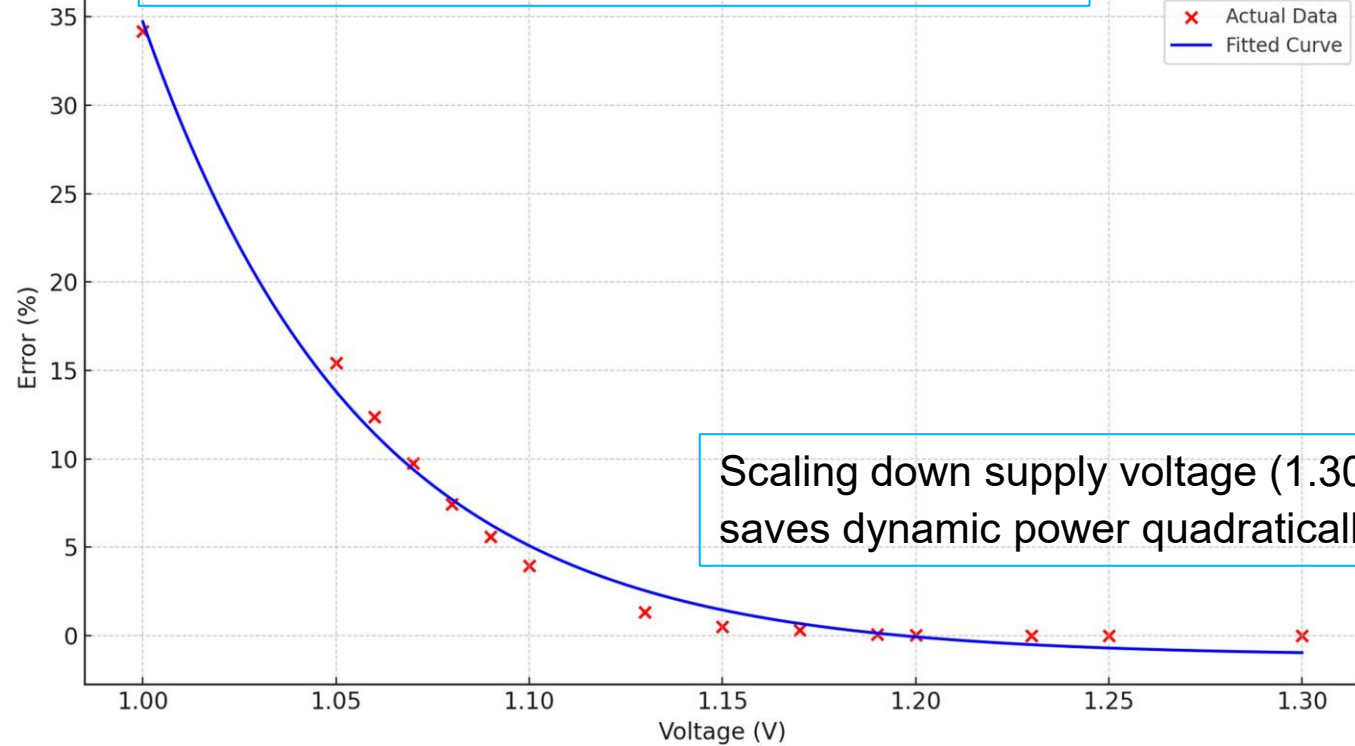


(a) Incoming Leakage path



(b) Outgoing Leakage paths

Trade-off: Triggers an exponential increase in memory retention errors (up to ~35% BER)



(c) Graph showing Voltage vs Error

Scaling down supply voltage (1.30V to 1.00V) saves dynamic power quadratically ($P \propto V_{DD}^2$)

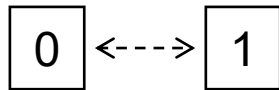
Memory Fault Behaviors Under Undervolting

Soft Errors (Bit-Flips):

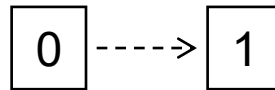
- Temporary data glitches
- Automatically fixed during next memory write cycle

Permanent Errors (Stuck-at Faults):

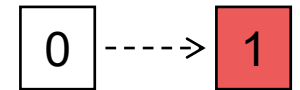
- Caused by manufacturing defects
- Exposed due to aggressive undervolting
- Weak cells leak charge, permanently locks physical cell to ground (0) or supply voltage (1)



(a) Normal bit update

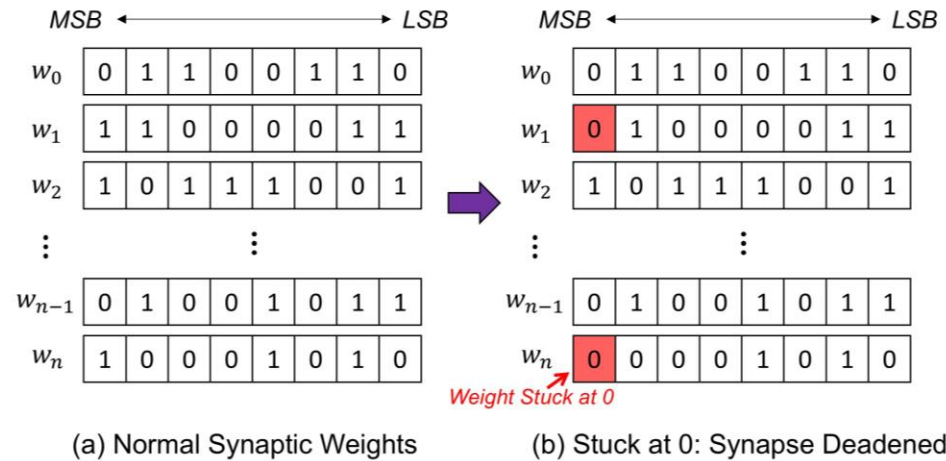


(b) Bit flip



(c) Bit stuck-at

Impact of SA0 vs SA1 Faults

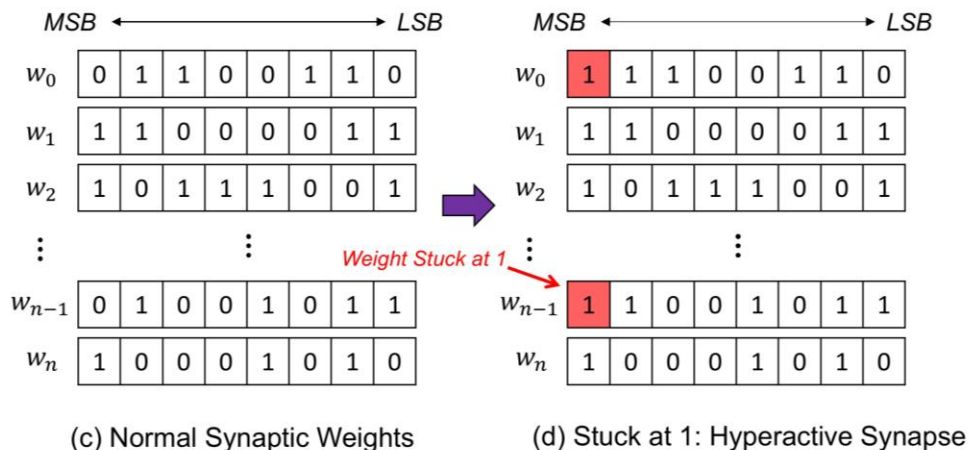


Stuck-at-0 (SA0):

- Weight locked to minimum conductance
- Easily absorbed, accuracy remains stable (>90%).

Stuck-at-1 (SA1)

- Weight locked to maximum conductance ($W=1.0$)
- Neuron fires continuously
- Suppresses all healthy neurons
- Accuracy collapses from **92.0% to 10.9%** *STDP Learning model 784 * 1000, MNIST dataset



(e) Normal Field



(f) SA0 Field



(g) SA1 Field

Problem Statement and Contributions

Problem:

- Near-threshold undervolting to lower consumption
- Low voltages cause stuck-faults and flip bits
- SA1 destroys STDP learning

Contributions:

- Characterized SA1 behavior under eDRAM voltage scaling
- Proposed **STDP-DIP** to identify, isolate and prune faulty neuron

Related Work

- Error Correction Codes ^[3]:
 - Requires 12.5% to 37.5% storage area overhead
 - Encoding/decoding negates 40% energy saved
- Checkpointing ^[4] :
 - Milliseconds of latency per save
 - Incurs >100% to 200% power overhead
- Inference Mitigations ^[5] :
 - Protects only pre-trained networks
 - Energy overhead 15% to 30%
- Memory Checking ^[6] :
 - Can check for MSB and LSB
 - Energy overhead 30% to 50%

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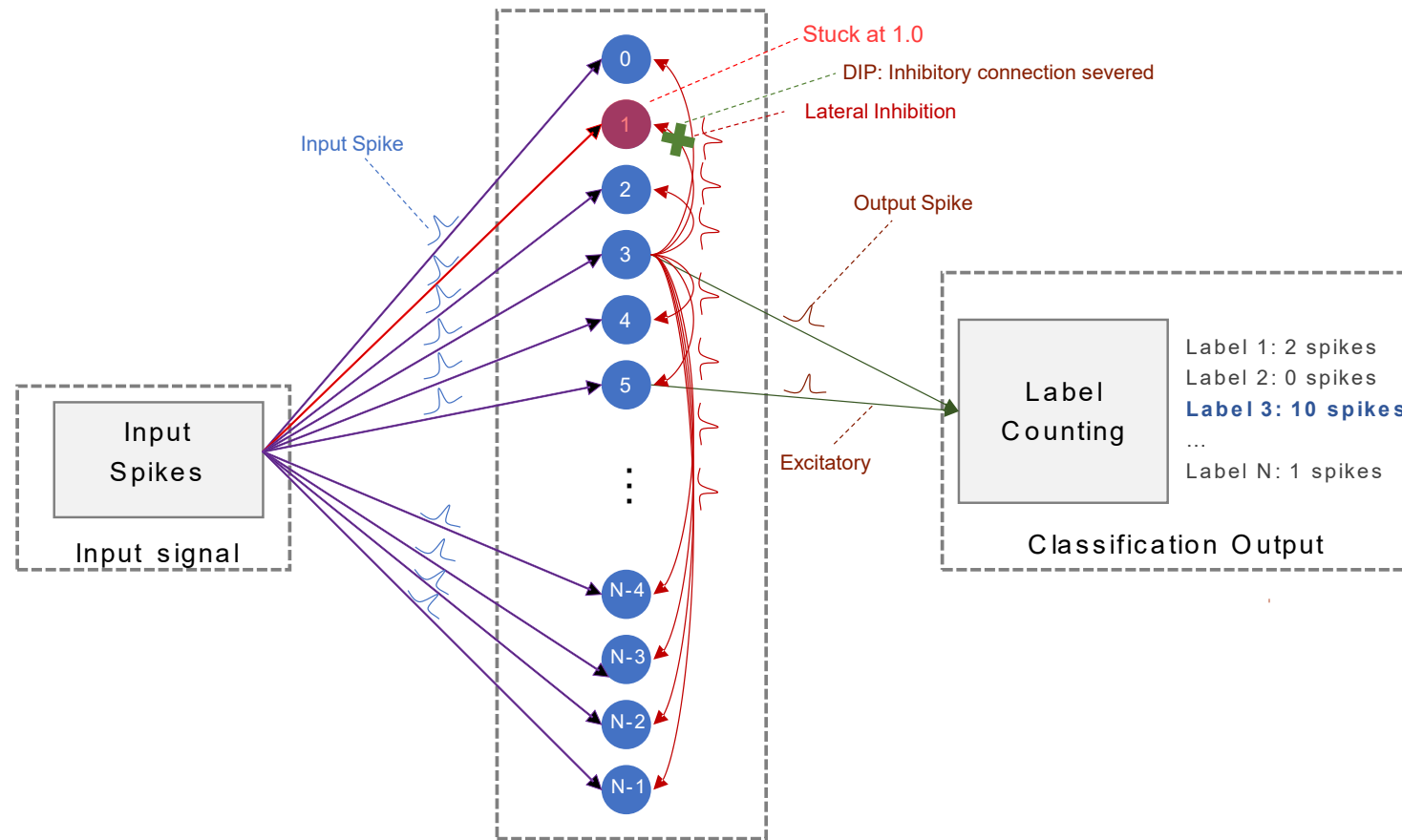
STDP-DIP: Diagnostic Isolation Pruning

Phase 1: Scans the initial weight matrix W_0

Phase 2: Faulty neuron is structurally severed

Phase 3: Output of faulty neuron unlabeled

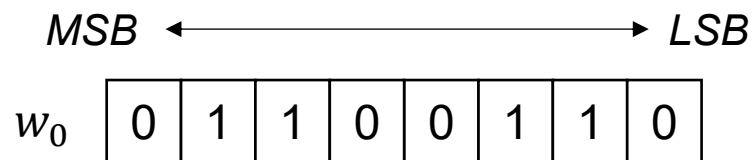
STDP-DIP: Proposal Overview



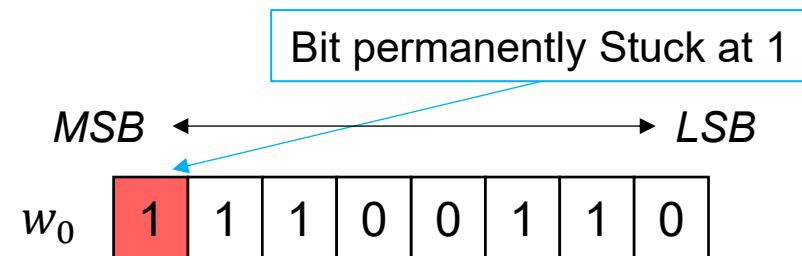
Overview of STDP network and STDP-DIP

STDP-DIP: Detection

- MSB faults cause most impact
- LSB faults cause minimal impact
- Detection based on our observation of bits
- Memory check exists but needs more energy and time
- ECC negates 40% energy saved

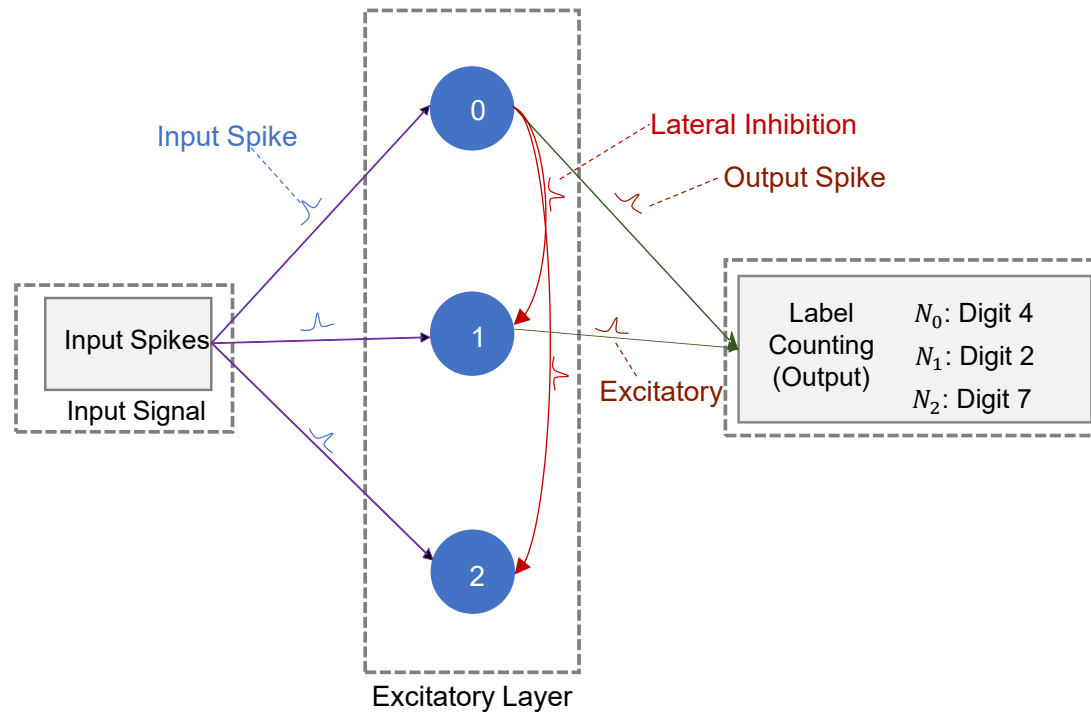


(a) Bit values of synaptic weights of healthy neuron



(b) Bit values of synaptic weights of SA1 neuron

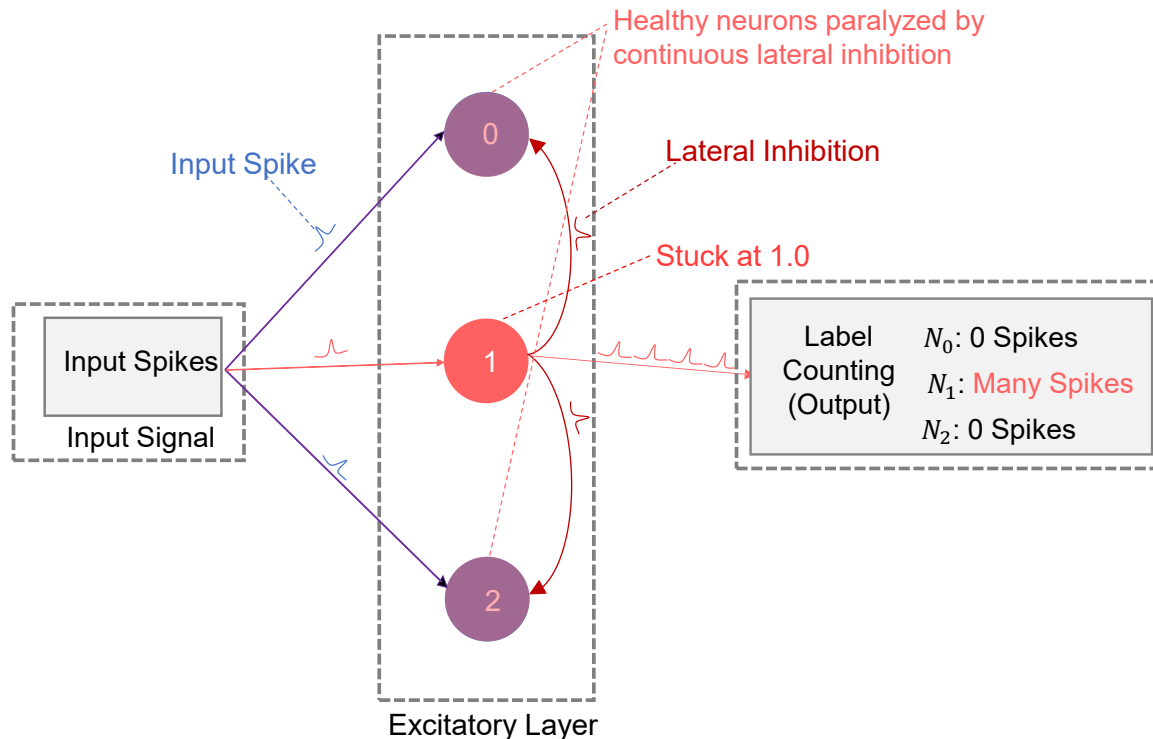
STDP-DIP: Algorithm (1)



Normal STDP Operation

- Brief lateral inhibition triggered by each spike
- Healthy competition and diverse feature learning

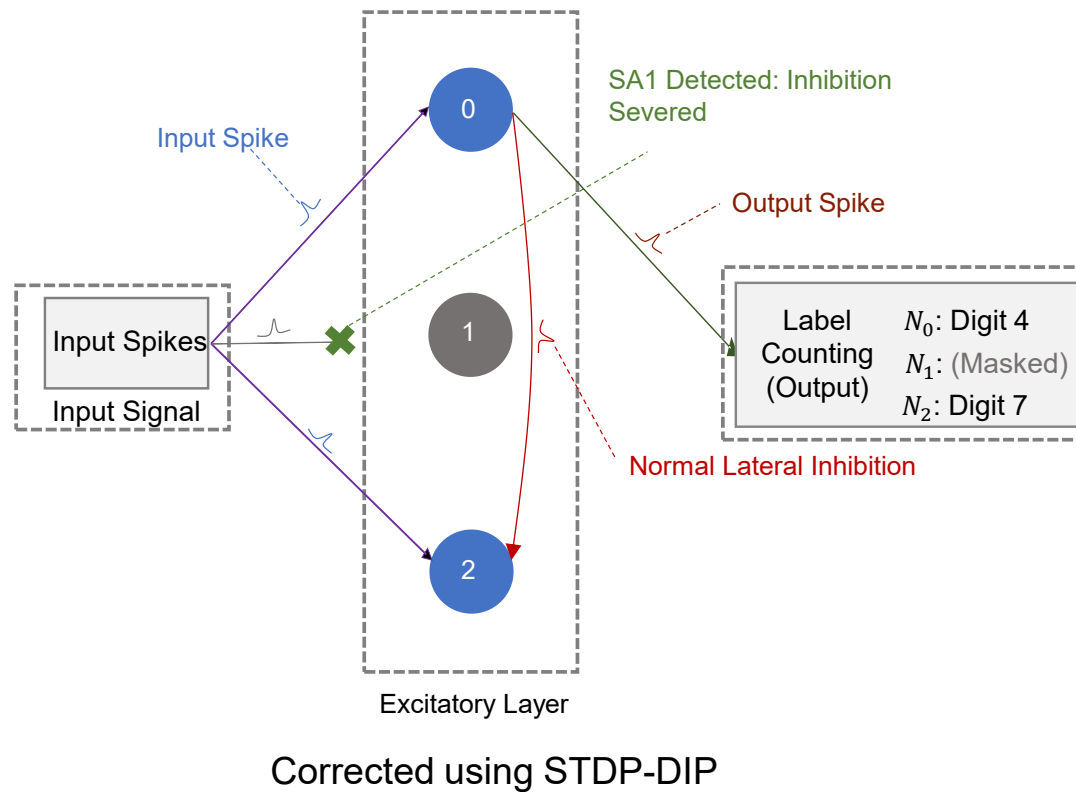
STDP-DIP: Algorithm (2)



Uncorrected SA1 Fault

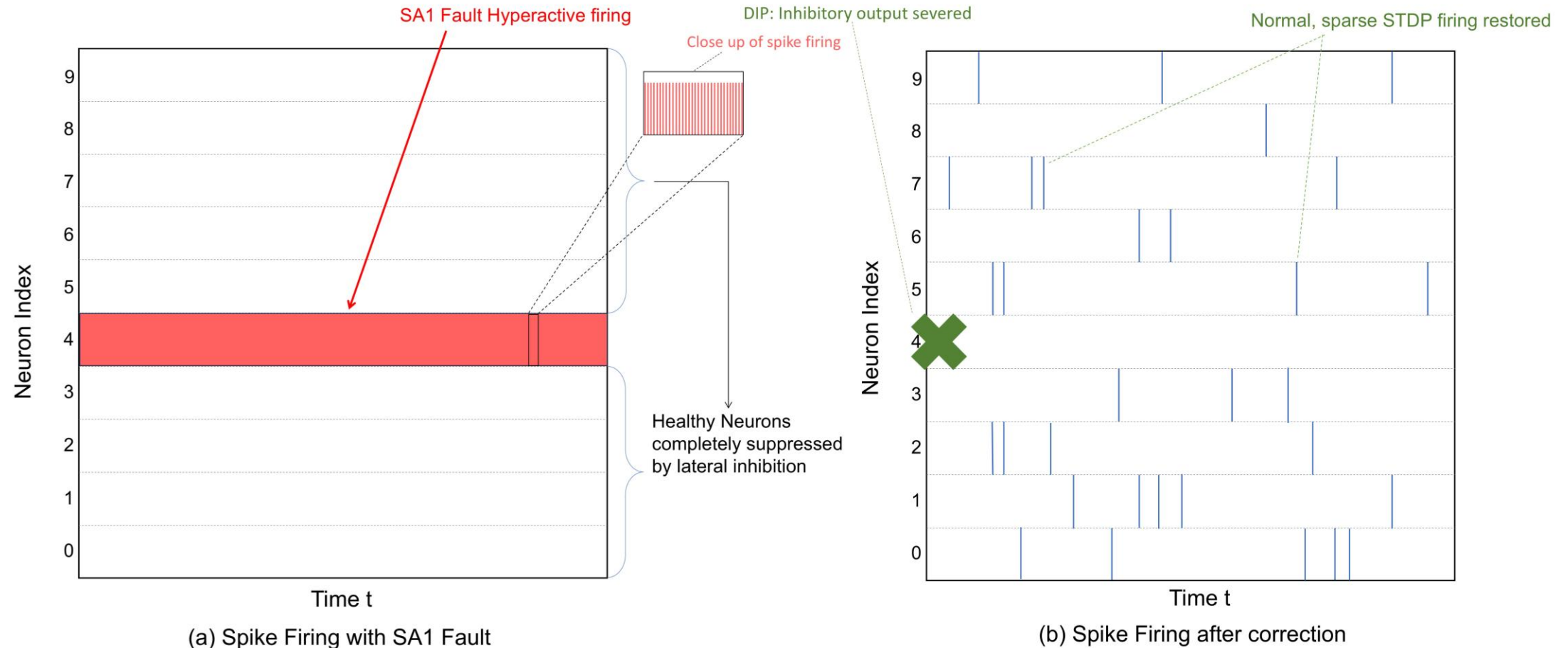
- SA1 fault causes neuron to be stuck-at-1
- Faulty neuron fires continuously
- Runaway lateral inhibition, paralyzes neurons, stops learning

STDP-DIP: Algorithm (3)



- Scans for faulty bit values
- Faulty neuron connection severed
- Healthy STDP spiking restored

Spike Behaviour Restoration



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Results: Simulation Setup

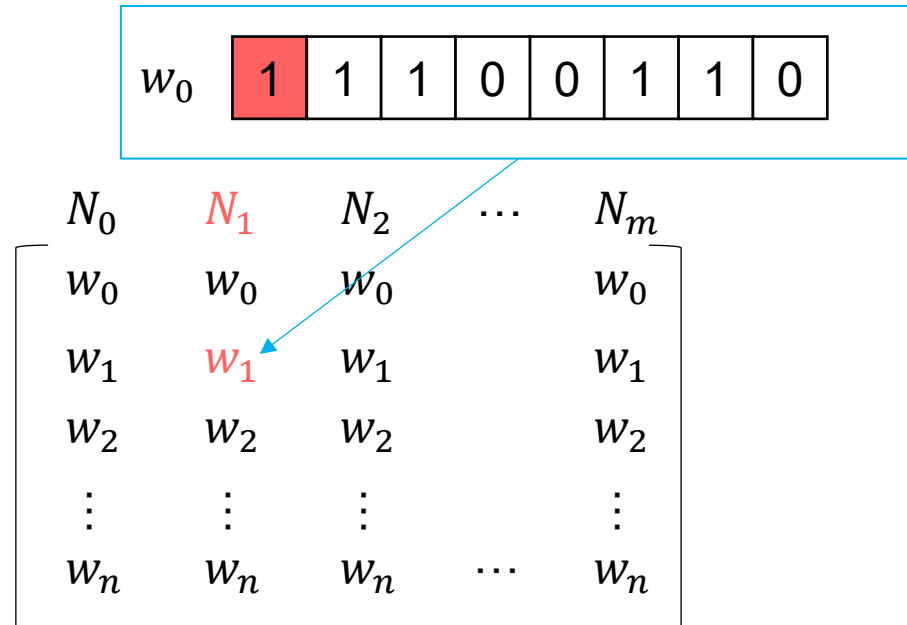
Environment: Python, PyTorch, BindsNET.

Hardware Model: 45nm PTM CMOS eDRAM

Network: LIF-STDP model

Dataset: Poisson-encoded MNIST.

Results: Fault Insertion

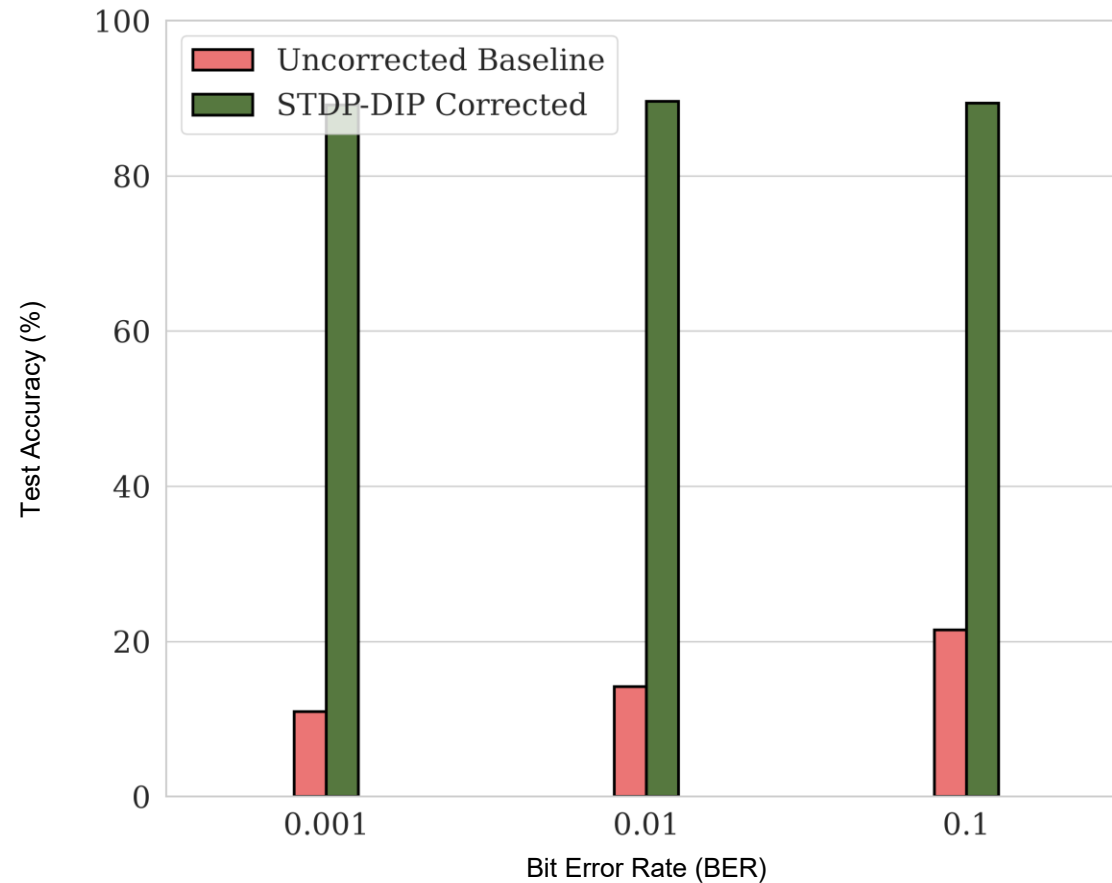


- Faulty neurons chosen randomly
- MSB of weights flipped manually
- Simulates SA1 fault

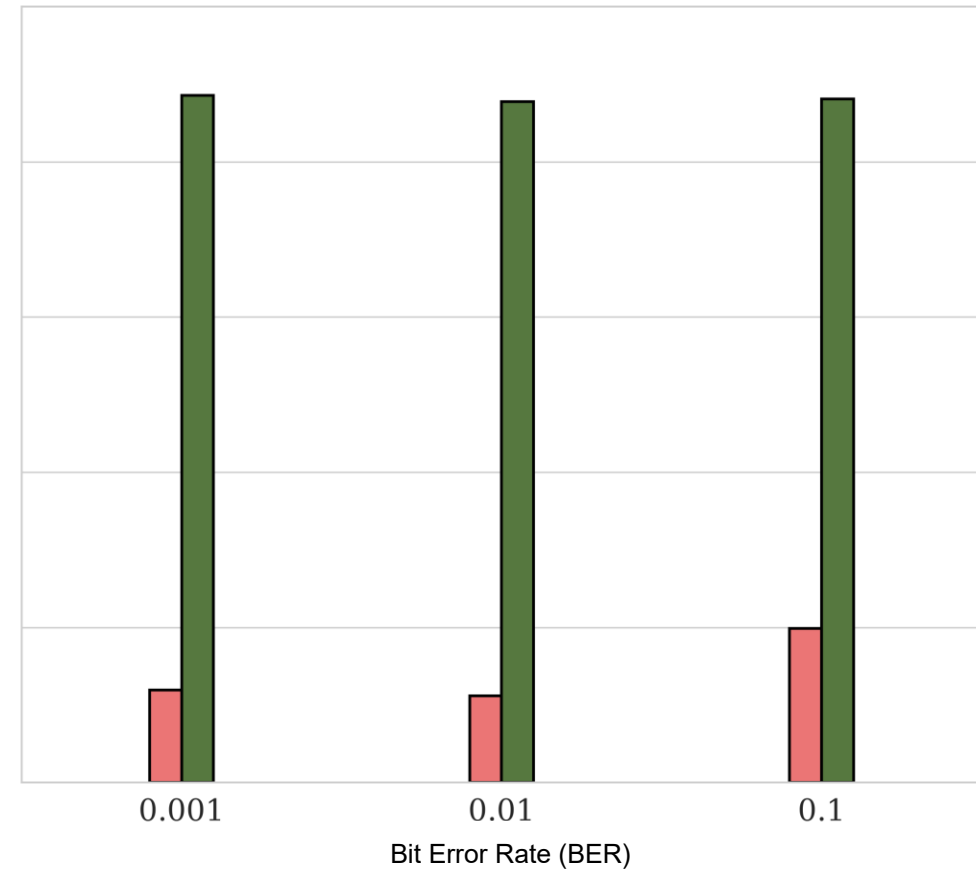
m : Number of Neurons (1000)

n : Number of weights (784)

Results: 400 Neurons



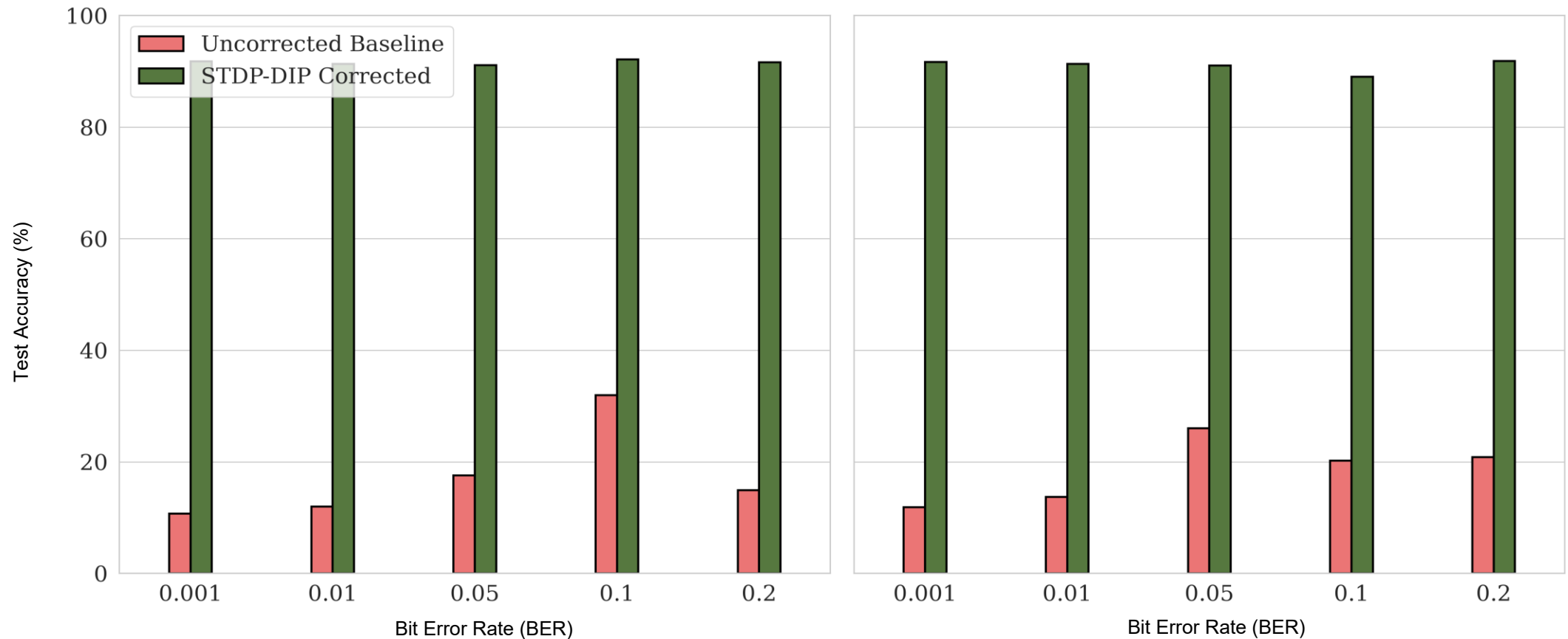
(a) Plot between accuracy before and after correction for 5 SA1 neurons in a 400 neuron network



(b) Plot between accuracy before and after correction for 15 SA1 neurons in a 400 neuron network

Accuracy improved by **73% - 79%**

Results: 1000 Neurons

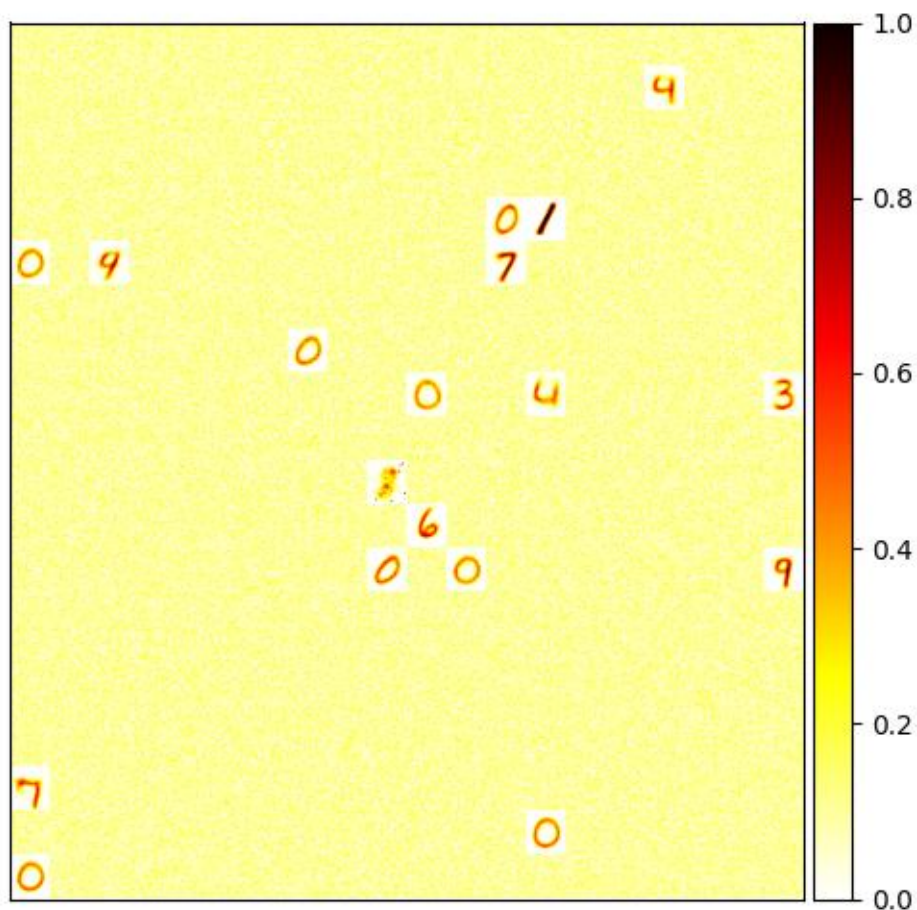


(a) Plot between accuracy before and after correction for 5 SA1 neurons in a 1000 neuron network

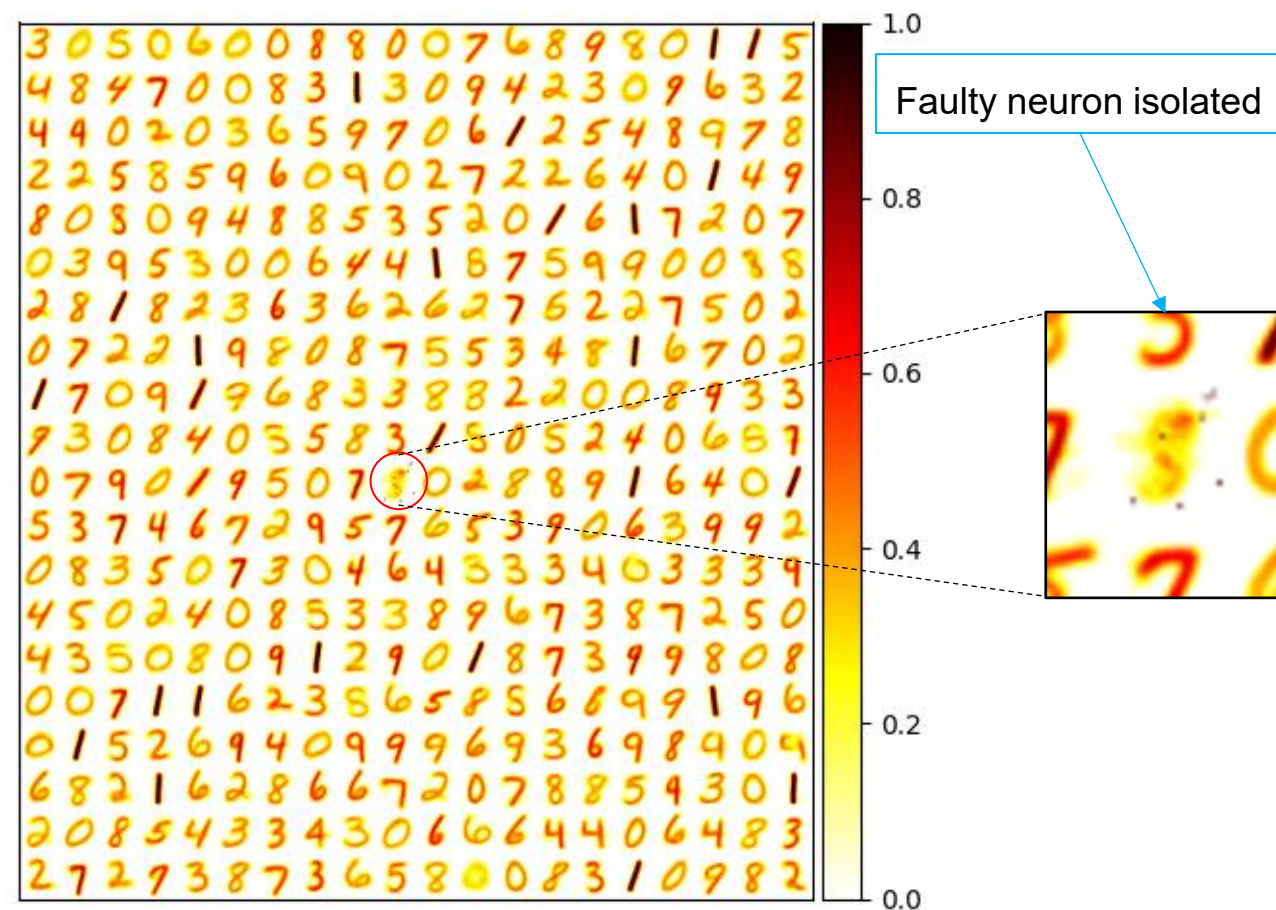
(b) Plot between accuracy before and after correction for 15 SA1 neurons in a 1000 neuron network

Accuracy improved by 71% - 81%

Results: Training With Fault Correction



(a) Network trained with SA1 fault and no correction



(b) Network trained using proposed STDP-DIP framework

Results: Energy Saved

$$E_{relative} = \left(\frac{V_{low}}{V_{nom}} \right)^2 = \left(\frac{1.00V}{1.30V} \right)^2 \approx 59.17\%$$

V_{low} : Undervolted potential

V_{nom} : Nominal potential

- 40.83% reduction by scaling supply voltage down to 1.00V
- 20.4% to 30.6% total energy reduction

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Conclusion:

- Identified SA1 memory faults
- Developed STDP-DIP to detect and prune faulty neurons
- Restored >91% accuracy while saving 30.6% system energy

Future Work:

- Scaling for deep convolutional SNNs and other neural networks
- Testing mixed-fault scenarios (SA0 + SA1) on physical chips

Reference

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Thank you for your attention!